Theoretical Analysis of Gas Turbine Inlet Foggers and their Effectiveness

William Sabol
Mechanical Engineer

Mott MacDonald, Inc.

Abstract

Inlet foggers are widely used in some areas to enhance gas turbine output, yet theoretical and practical knowledge pertaining to fogging technology is not well documented. The advent of boiler feedwater fogging has sparked questions regarding its thermodynamic and operational implications over traditional fogging techniques. This paper aims to improve the theoretical understanding, provide a practical tool to evaluate fogger effectiveness, and shed some light on the thermodynamic effects of using warm boiler feedwater over standard ambient temperature demineralized water. Mott MacDonald developed a model of the thermodynamic interaction between a water droplet and the surrounding humid air. Both convective and latent heat transfers are considered in this model as well as the temperature dependency of air and water properties. Given ambient atmospheric conditions, this model determines the final air properties as it enters the compressor. It can be used to evaluate the design parameters of new systems and diagnose problems with existing ones. An example is presented showing the model predictions and validation against test data. The increase in inlet water temperature showed a slight rise in final air temperature.

Acknowledgements: McIan Amos, Joe Funk, Steve Kemper, Stuart Sabol

Introduction and Background

Mott MacDonald began surveying the inlet fogging systems at several combined cycle power plants in 2011, most of which were standard in design. Several pumps supplied demineralized water in increasing amounts to a nozzle array in the filter house depending on wet bulb temperature and ambient temperature. Each pump each ran at a single speed; therefore, supplying additional water meant running more pumps.

In general, these systems come complete with a slew of problems. The fogging pumps are not designed to pump demineralized water. Demineralized water's low lubricity will degrade and wear pumps not designed for the task. Without constant rebuilding, degraded pumps can vibrate violently, crack tubing, and send debris upstream that clog filters and nozzles.

One facility Mott MacDonald surveyed, Plant A, redesigned their traditional system in an effort to eliminate these problems. Their system utilizes their boiler feedwater direct from the boiler feedwater pump discharge. They tapped their boiler feedwater line, ran it through a heat exchanger to cool the water to approximately 200 °F, and then sent it to their existing fogging skid. Instead of turning multiple pumps on and off to control flow, Plant A split the line and installed control valves that open and close, mimicking the operation of the pumps.

Plant A has four gas turbine units at their facility, each with a fogging skid. Before installing their new boiler feedwater fogging system on all their units, a single test unit was established. Once installed, they noticed that the turbine with the new fogging system was generating 2-3 MW more than the other units. The operators decided that the temperature of the fogging water was responsible for the power spike because that was the only parameter that was different. Their traditional foggers were using water at ambient temperature, and the new system was using water at 200 °F.

This 2-3 MW spike is a perplexing phenomenon and poses a challenge to understand the thermodynamics of fogging operations. Mott MacDonald decided to model this thermodynamic system and develop a model to study and evaluate fogging systems. Among other things, this model can help us determine the water temperature's role in the 2-3 MW spike seen at Plant A.

This paper develops an analytical model that describes the convective and latent heat transfer between the water droplets and air while in the filter house and compares the resultant air properties when water temperatures are less than, equal to, and greater than the initial air temperature. Then the model simulates scenarios given actual plant data to verify how well this model reflects actual fogging systems in the field.

Evaporation Kinetics

Evaporation is vaporization of a liquid from its surface. Evaporative mass transfer is analogous to convective heat transfer. It can occur naturally or be forced if the air around the liquid is moving relative to the surface.

Evaporation is not boiling. Boiling requires an energy source to raise the liquid temperature to the boiling point. At the boiling point, rather than increasing the liquid temperature, the energy flux transfers packets of water from the liquid state to the vapor state.

When a liquid evaporates, molecules diffuse from the surface. At the surface of any liquid is a thin layer where the molecules have more energy and the physical state is somewhere between liquid and vapor. This is the Knudsen layer [1][6]. Depending on the surface tension and the vapor concentration gradient, molecules from the Knudsen layer may diffuse into the air naturally or forcibly [12]. As high energy molecules in Knudsen layer diffuse away from the surface, the total energy of the liquid, and therefore temperature, decrease.

The process described above is evaporative cooling. Heat from the surroundings can reheat the liquid, replenish the Knudsen layer, and allow more liquid to evaporate. The surroundings will continually cool until all the liquid evaporates or until the surroundings reach saturation.

Complexities and Assumptions Related to the Droplet Model

Molecular behavior can be difficult to measure and describe mathematically. Much of what happens on the molecular level is probabilistic and often impossible to predict accurately without quantum concepts. On the macroscopic level, the classical theory is mostly empirical and usually based on best-fit curves and a host of dimensionless numbers, each with their own set of caveats.

While inlet foggers are widely accepted tools for increasing gas turbine output in certain areas, theoretical and practical knowledge pertaining to fogging technology is not well documented [11]. Prior to the work of Meher-Homji, C.B., et al, “not a single technical paper existed covering the physics and engineering of the fogging process” [11]. The scarcity of information on this subject leaves much to be desired, particularly concerning the theoretical model described in the next section.

The following theoretical model was designed under a set of assumptions to lighten technical and computational burdens. This model makes the following assumptions.

- *Droplets are distributed perfectly in the air upon exiting the nozzle*. In practice, the concentration of droplets is not constant over a filter house cross section. Therefore, some volumes of air will reach saturation before others, slowing evaporation until that volume mixes or diffuses with unsaturated air.
- *Droplets exit the nozzle at the same velocity as the air*. In reality, for a very short time after exiting the nozzle, there are differences between the air velocity and droplet velocity, meaning there is forced heat transfer. However, due to their size and low Reynolds number (< 1), droplets attain air stream velocity within a few milliseconds. The difference between natural and forced evaporation over that short time is negligible.

- *Droplets are perfectly spherical.* This is a common assumption. Depending on turbulences and differences between the droplet and the air, the droplet may have some eccentricity, but those fluctuations are negligible.
- *The droplet and air system are thermodynamically isolated from the filter house and its surroundings*. Heat transfer between the air and the filter house structure is negligible. In reality, on a hot summer day, the filter house walls will be hotter than the air inside. Convection between them will offset evaporative cooling. This is beyond the scope of this project.
- *The residence time within the gas turbine filter house is 1.5 seconds*. Residence times are dependent on air stream velocity and filter house design but most range from 1-2 seconds [11].
- *A Lumped Capacitance model is valid for the droplet and air*. Lumped capacitance means one can assume a constant mass or temperature gradient within a body. This assumption allows the model to focus on the interaction between the droplet and its surroundings rather than having to consider convection and diffusion currents within the droplet and the air.
- *Air stream is not turbulent*. Droplets in turbulent surroundings experience much faster heat and mass transfer rates. In reality, there is some turbulence in the air around support posts and other irregularities.
- *Air is an ideal gas*. This is a very common assumption, albeit impossible for any real gas. Water vapor is not as ideal as air, but with the low water vapor concentrations in these simulations, the inaccuracies are negligible. This model uses the ideal gas law in some instances to determine some properties of humid air. Other properties are determined by empirical or psychrometric means.

Droplet Heat and Mass Transfer Model

Lumped Capacitance Validation

The Biot Number is used to validate the lumped capacitance assumption listed above. A Biot number below approximately 0.1 implies approximately 5% error in assuming a constant temperature or concentration profile, thus it is reasonable to assume lumped capacitance [9]. The Biot numbers for the droplet, the air, and the mass Biot number for the air are defined as

$$Bi_a = \frac{h_{cv} D_d}{6k_a}$$

$$Bi_d = \frac{h_{cv} D_d}{6k_w}$$

$$Bi_m = \frac{K_{mass} D_d}{6\delta_a}$$

The definitions and derivations of h_{cv}, K_{mass}, δ_a, and k_a are explained in detail in the following subsection. However, it is important to establish the validity of lumped capacitance before moving forward. While running the simulation, the model generated real time Biot numbers for each simulation. These numbers, shown below, are acceptable for this application.

$$Bi_a \approx 0.3$$
$$Bi_d \approx 0.02$$
$$Bi_m \approx 0.3$$

Psychrometrics

The vapor pressure, P_{vap}, is the partial pressure of water vapor in humid air. The saturated vapor pressure, P_{sat}, is the partial pressure of water vapor at saturation. Relative humidity is the ratio of vapor pressure to saturated vapor pressure.

$$RH = \frac{P_{vap}}{P_{sat}}$$

The Arden Buck equation for saturated vapor pressure at a given temperature (Kelvin) as defined in [2] as

$$P_{sat} = 611.21 * e^{\left(19.8428 - \frac{T_a}{234.5}\right)\left(\frac{T_a - 273.15}{T_a - 16.01}\right)}$$

Define the partial pressure of dry air with the ideal gas law

$$P_{da} = \frac{\rho_{da} R T_a}{M_{da}}$$

where ρ_{da} is the density of dry air derived using a linear regression of the data shown in [8]

$$\rho_{da} = 2.3723 - 0.00397 * T_a$$

Therefore, the total pressure of the air is

$$P_{tot} = P_{vap} + P_{da}$$

Specific humidity is the water vapor to air mass ratio.

$$SH = \frac{m_v}{m_{da}}$$

We can also write an equation for specific humidity in terms of the water vapor and dry air mass concentrations. Note that the mass of dry air is constant, but its concentration is time dependent.

$$SH = \frac{c_v}{c_{da}}$$

where

$$c_v + c_{da} = 1$$

From these two definitions for specific humidity, we can determine the mass concentrations of water vapor and dry air. These are used to determine the mass of dry air and the density of humid air.

$$m_{da} = c_{da} m_a$$
$$\rho = c_v \rho_{wv} + c_{da} \rho_{da}$$

where ρ_{wv}, the density of water vapor, is defined using the ideal gas law

$$\rho_{wv} = \frac{P_{vap} M_{water}}{R T_a}$$

The concentration of water vapor is low in comparison to the concentration of dry air, thus the inaccuracies due to the ideal gas law are negligible.

Heat and Mass Transfer Equations

Commencing with heat balances of the droplet and air, thermodynamic equilibrium between the droplet and air occurs when the partial pressure of water vapor is equal to the saturated vapor pressure and when the air temperature is equal to the droplet temperature.

$$\dot{Q}_d(t) = \dot{Q}_{conv}(t) + \dot{Q}_{lat}(t)$$

$$\dot{Q}_a(t) = -\dot{Q}_{conv}(t)$$

The full differential heat rates for the droplet and air describe energy changes when mass, heat capacity, and temperature change.

$$\dot{Q}_d(t) = \dot{m}_d C_d T_d + m_d \dot{C}_d T_d + m_d C_d \dot{T}_d$$

$$\dot{Q}_a(t) = \dot{m}_a C_a T_d + m_a \dot{C}_a T_a + m_a C_a \dot{T}_a$$

Combining the equations for heat rate and the heat balance give

$$\dot{m}_d C_d T_d + m_d \dot{C}_d T_d + m_d C_d \dot{T}_d = \dot{Q}_{conv}(t) + \dot{Q}_{lat}(t)$$

$$\dot{m}_a C_a T_d + m_a \dot{C}_a T_a + m_a C_a \dot{T}_a = -\dot{Q}_{conv}(t)$$

Convective Heat Transfer

The convective heat rate is defined by Newton's Law of Cooling as

$$\dot{Q}_{conv} = h_{cv} S_d (T_a - T_d)$$

The convective heat transfer coefficient, h_{cv}, is derived from the Nusselt number

$$Nu = \frac{h_{cv} D_d}{k_a}$$

where the thermal conductivity of air, k_a, is defined by [11] as

$$k_a = (46.766 + 0.7143 * T_a) * 10^{-4}$$

The Nusselt number is defined empirically for different situations. For natural convection around a sphere, it is defined as a function of the Prandtl and thermal Grashof numbers in [11] as

$$Nu = 2 + 0.6 * {Gr_t}^{0.25} * Pr^{0.33}$$

where the thermal Grashof number is defined by

$$Gr_t = \frac{\rho^2 g \beta |T_a - T_d| {D_d}^3}{{\mu_a}^2}$$

μ_a is the dynamic viscosity of air can be derived from Sutherland's formula [13]

$$\mu_a = 1.512 * 10^{-6} * \frac{{T_a}^{3/2}}{T_a + 120}$$

The thermal dilation coefficient, β, is defined as, and assuming air is an ideal gas, evaluates to

$$\beta = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T_a}\right) = \frac{1}{T_a}$$

The Prandtl number, given by

$$Pr = \frac{\mu_a C_a}{k_a}$$

C_a is the specific heat of humid air given as the weighted average of the specific heats of water vapor and dry air by their respective concentrations.

$$C_a = c_{da} C_{da} + c_v C_{wv}$$

where C_{da} and C_{wv} are the specific heats of dry air and water vapor are derived from data shown in [8] and [4] respectively

$$C_{da} = 0.000402 * {T_a}^2 - 0.2026 * T_a + 1030.9$$
$$C_{wv} = 1772 + 0.312 * T_a$$

From the equations above, we can write the derivative of C_a with respect to time

$$\dot{C}_a = \dot{c}_{da}C_{da} + c_{da}\dot{C}_{da} + \dot{c}_v C_{wv} + c_v \dot{C}_{wv}$$

$\dot{c}_v$ and $\dot{c}_{da}$ are defined from the definitions of SH

$$\dot{c}_{da} = -\dot{c}_v$$

$$\dot{c}_v = \frac{\dot{m}_a}{m_a} c_{da}$$

Using that and the derivatives of C_{da} and C_{wv}, the above equation simplifies to

$$\dot{C}_a = \left(\frac{\partial C_{da}}{\partial T_a} c_{da} + \frac{\partial C_{wv}}{\partial T_a} c_v\right)\dot{T}_a + (C_{wv} - C_{da})\dot{c}_v$$

Latent Heat and Mass Transfer

The latent heat rate is defined a function of the mass exchange rate and the latent heat of vaporization.

$$\dot{Q}_{lat} = L_v \dot{m}_d$$

The latent heat of vaporization for water, L_v, can be derived from the Clausius-Clapeyron relation that, for evaporation, states

$$\frac{dP_{sat}}{dT} = \frac{L_v}{T\Delta v}$$

where Δv is the change in specific volume from the liquid state to the vapor state. Using the ideal gas law, Δv is defined as

$$\Delta v = \frac{RT}{M_w P_{sat}} - \frac{1}{\rho_w}$$

where ρ_w is the density of water at a given temperature derived from a linear regression of the data found in [7]

$$\rho_w = 786.5 + 1.681 * T_d - 0.003272 * {T_d}^2$$

The latent heat of vaporization evaluated at the water temperature finally comes to

$$L_v = P_{sat} T_d \left(\frac{RT_d}{M_w P_{sat}} - \frac{1}{\rho_w}\right)\left(\frac{1192134 + 32.02 * T_d - {T_d}^2}{234.5 * (T_d - 16.01)^2}\right)$$

Because mass exchange happens at the water's surface, it is dependent on the surface area exposed to the air, S_d. We can define mass rate as a function of a mass flux, Φ_{evap}. Let a positive mass flux mean an increase in droplet mass.

$$\dot{m}_d = S_d \Phi_{evap}$$

When the air is not saturated, water will evaporate. We can model evaporation as the diffusion of water vapor into air. The driving force for diffusion is the difference in the water vapor concentration between the droplet's surface and the air. Mass flux is defined as

$$\Phi_{evap} = K_{mass} * \Delta C_{\mathrm{mass}}$$

In accordance to the statement above, the driving force, ΔC_{mass}, is defined as

$$\Delta C_{\mathrm{mass}} = \rho_{wv} - \rho_{knd}$$

Use the ideal gas law to define ρ_{knd} as

$$\rho_{knd} = \frac{M_{water} P_{knd}}{R T_d}$$

where M_{water} is the molar mass of water, R is the universal gas constant, and P_{knd} is the equivalent saturation pressure of the droplet's surface. P_{knd} is dependent on the ambient saturation pressure and the surface tension of the water defined in [12] as

$$P_{knd} = P_{sat} e^{\left(\frac{4\sigma M_{water}}{\rho_w R T_d D_d}\right)}$$

where σ is the surface tension of the water defined in [12] as

$$\sigma = 2.1 * 10^{-7} * \left(\frac{\rho_w}{M_{water}}\right)^{\frac{2}{3}} (T_C - T_d)$$

where T_C is the critical temperature of water.

The mass diffusion coefficient is given by

$$K_{mass} = \frac{Sh * \delta_a}{D_d}$$

where Sh is the Sherwood number and δ_a is the mass diffusivity for air defined in [11] as

$$\delta_a = 2.26 * 10^{-5} * \left(\frac{101325 * T_a}{273.15 * P_{tot}}\right)$$

The Sherwood number, defined in [11], is a function of the mass Grashof number and the Schmidt number

$$Sh = 2 + 0.6 * {Gr_m}^{0.25} * Sc^{0.33}$$

The Schmidt number is defined as

$$Sc = \frac{\mu_a}{\rho \delta_a}$$

and the mass Grashof number is defined by

$$Gr_m = \frac{\rho^2 g \beta_m |\rho_{wv} - \rho_{knd}| {D_d}^3}{{\mu_a}^2}$$

where β_m is defined as

$$\beta_m = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial \rho_{wv}}\right)$$

From the *Psychrometrics* section, we defined an expression for ρ in terms of the densities of water vapor and dry air. From this, we can evaluate β_m

$$\beta_m = \frac{c_v}{\rho}$$

Final Differential Equations

From the equations above, differential equations for the mass of the droplet and the surrounding air are

$$\dot{m}_d = S_d \Phi_{evap}$$
$$\dot{m}_a = -S_d \Phi_{evap}$$

The differential equations for the droplet and air temperature are

$$\dot{T}_d = \frac{S_d\left(h_{cv}(T_a - T_d) + L_v \Phi_{evap}\right) - \dot{m}_d C_d T_d}{m_d \left(C_d + T_d \frac{\partial C_d}{\partial T_d}\right)}$$
$$\dot{T}_a = \frac{h_{cv} S_d (T_d - T_a) - \dot{m}_a T_a \left(C_a + c_{da}(C_{wv} - C_{da})\right)}{m_a \left(C_a + T_a \left(\frac{\partial C_a}{\partial T_a} c_{da} + \frac{\partial C_{wv}}{\partial T_a} c_v\right)\right)}$$

where C_d is the specific heat of water defined using a linear regression of the data shown in [7]

$$C_d = -0.000099 * {T_d}^3 + 0.10913 * {T_d}^2 - 39.178 * T_d + 8785.4$$

Computational Simulation

The computational simulation works in an iterative manner to numerically integrate the differential equations derived above and provide the transient behavior of the fogger state space. We define the state space vector based of the differential equations derived above.

$$\vartheta = (T_d \quad T_a \quad m_d \quad m_a)^T$$

Differentiating the state space vector allows one to use the equations derived above and integrate the space from a set of initial conditions in a standard left Riemann sum fashion.

$$\vartheta\big((i+1)\Delta t\big) = \vartheta(i\Delta t) + \Delta t * \dot{\vartheta}(i\Delta t)$$

where $i \geq 0$ represents the iteration number and Δt is the time step.

Initial Conditions

For any initial conditions, the model simulates the system's behavior when governed by those differential equations. The required initial conditions are those in the state space vector: initial air and water temperatures and initial masses for the droplet and surrounding air.

The initial air and water temperatures are directly hard coded. The initial mass of the air and droplet come from the following relationship. Because we assumed perfectly distributed droplets, the flow ratio is proportional to the mass ratio of a single droplet to its portion of surrounding air. If we prescribe an initial droplet diameter, D_d, then

$$m_d(0) = \frac{1}{6}\rho_w \pi {D_d}^3$$
$$m_a(0) = m_d(0)\frac{\epsilon_a l}{\epsilon_w}$$

where ϵ_a is the standard air flow through the turbine, ϵ_w is the flow of water from the fogging skid, and l is the turbine load.

The initial composition of the air is derived using an initial relative humidity and the steps presented in the *Psychrometrics* subsection to find the concentration of dry air and water vapor.

If a wet bulb temperature is prescribed rather than relative humidity, the relative humidity is calculated. The model first calculates the saturated vapor pressure at the wet bulb temperature,

P_{wb}, and saturated vapor pressure at the initial air temperature using the Arden Buck equation. The model then calculates the vapor pressure with

$$P_{vap} = P_{wb} - 101325 * (T_a - T_{wb}) * 0.00066 * \left(1 + 0.00115 * (T_{wb} - 273.15)\right)$$

Now it is possible to calculate relative humidity as the ratio of the vapor pressure to the saturated vapor pressure.

Exit Conditions

Fogging stops when the system reaches one of these two conditions: the droplets exit the filter house, or the droplets evaporate completely. The former is an application of the assumption that the residence time in the filter house is 1.5 seconds. The latter condition depends upon how droplets behave at small sizes.

As droplet size decreases to the point where the droplet and air can no longer be regarded as discrete fluid continua, the classical theory presented previously does not apply [12][14]. This depends on the Knudsen number, defined as the ratio of the mean free path of a gas to the droplet radius [10][12].

$$Kn = \frac{2\lambda}{D_d}$$

where λ is the mean free path of water vapor defined in [7] as

$$\lambda = \frac{kT_d}{\sqrt{2}\pi {d_w}^2 P_{tot}}$$

where k is Boltzmann's constant and d_w is the molecular diameter of water. The theory presented in the previous section is only applicable for Knudsen numbers less than about 0.01 [14]. When Knudsen numbers are greater than 0.01, molecular interaction diminishes to the point where there is no molecular interaction, and discrete particle models are required to model the flow accurately [10][14].

As might be expected, instability is observed as the computation simulation approaches Knudsen numbers greater than 0.01. High Knudsen number flows are beyond the scope of this project, therefore let us consider the droplet to evaporate completely prior to the simulation becoming unstable, when $Kn \approx 0.08$. This corresponds to droplet diameters of about 2.8 microns.

The final exit condition for the simulation will be when

$$Kn > 0.08 \text{ OR } t > 1.5$$

The model returns values after the simulation ends. Based on the returned values, we can determine the effect of initial conditions on the system. The model returns values for the final air temperature, the final droplet diameter, the final relative humidity, the duration of the fogging process, and the percentage change in air density.

Model Results

This computational model allows a comparison of varying initial conditions on the system to analyze the transient behavior of states related to droplet behavior. Running the simulation using the ambient air conditions provided in the table below will show us the water temperature's effect on the system.

Table 1. Simulation Initial Conditions

Property	*Value*
Air Temperature	80 °F
Relative Humidity	50%
Water Flow Rate	60 gpm
Turbine Load	100%
Droplet Diameter	30 µm

We can consider three cases where the water temperature varies with respect to the air temperature. The first case will use an initial water temperature less than the air temperature at 60 °F. The second case will use an initial water temperature equal to the air temperature at 80 °F. The final case will use an initial water temperature greater than the air temperature at 140 °F.

Case I: $T_d < T_a$

Figure 1 shows the behavior of the system when the initial water temperature is 60 °F.

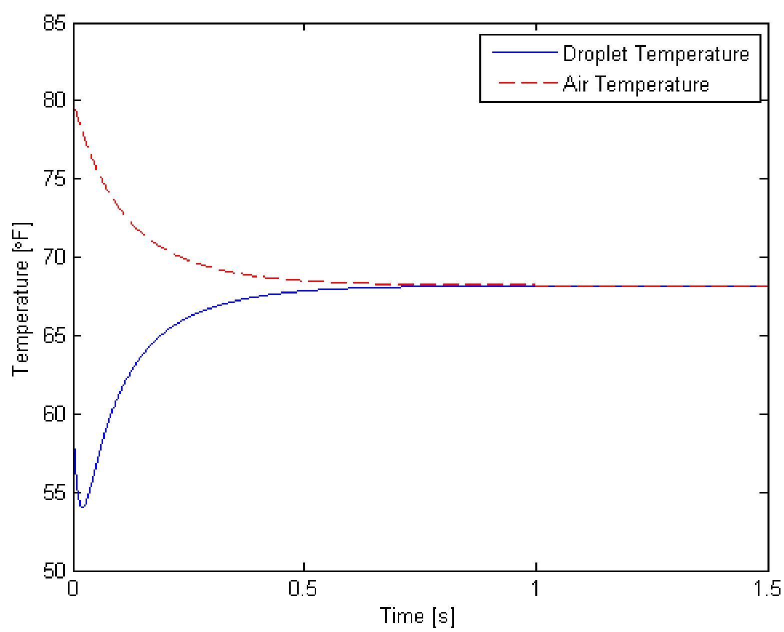

Figure 1. Transient behavior of 80°F air and a 60°F droplet at 50% RH

Figure 2 presents a graph of $\dot{Q}_{conv}$ and $\dot{Q}_{lat}$ to show how convection and latent heat transfer contribute to the temperature changes taking place in the droplet. In conjunction with Figure 1, it can explain how heat and mass transfer between the droplet and air.

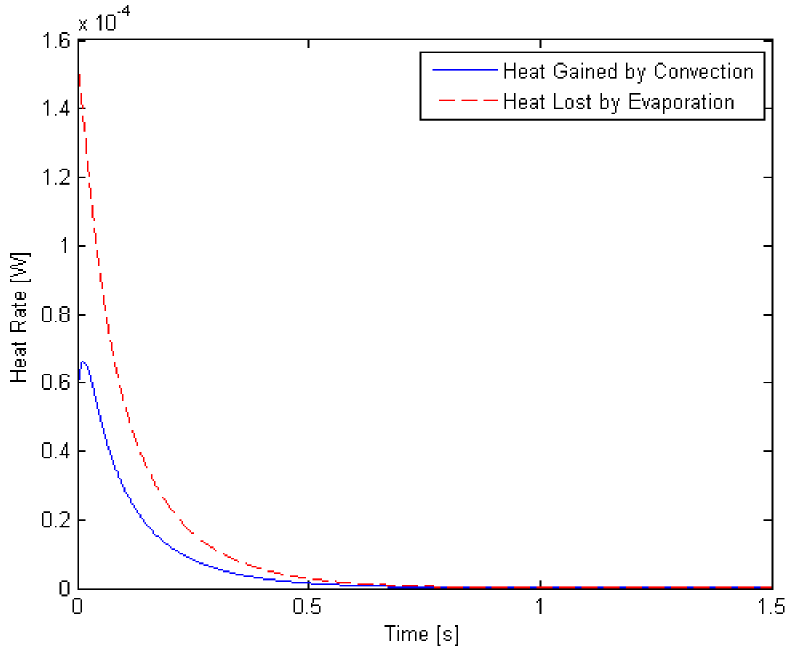

Figure 2. Contributions by convection and latent heat transfer on a droplet starting at 60°F

As the droplet exits the nozzle, latent heat transfer dominates because of the low water vapor concentration in the air. The water from the droplet evaporates, droplet temperature falls, and relative humidity increases, which slow evaporation. Meanwhile, convective heat transfer is rapidly increasing because of the droplet's temperature drop. Relative humidity and vapor pressure increase enough to bring the total heat transfer to a quasi-equilibrium state where latent heat transfer is proportional to convection. The latent heat transfer is approximately twice the magnitude of convective heat transfer, both decreasing in this manner until the end of the simulation.

Throughout this explanation, we have been studying the droplet's relationship to heat transfer. Keep in mind that throughout all this time, convective and latent heat transfer have been working together to bring the air temperature down. In this case, there is no heat transferred to the air.

Case II: $T_d = T_a$

Figure 3 shows the behavior of the system when the initial water temperature is 80 °F, the same temperature as the surrounding air.

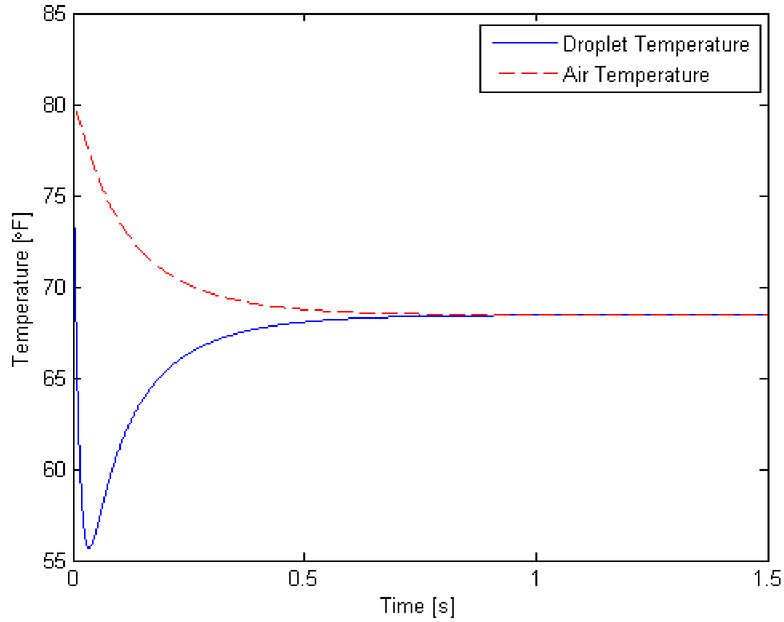

Figure 3. Transient behavior of 80°F air and an 80°F droplet at 50% RH

Figure 4 presents a graph of $\dot{Q}_{conv}$ and $\dot{Q}_{lat}$ to show how convection and latent heat transfer contribute to the temperature changes taking place in the droplet. In conjunction with Figure 3, it can explain how heat and mass transfer between the droplet and air.

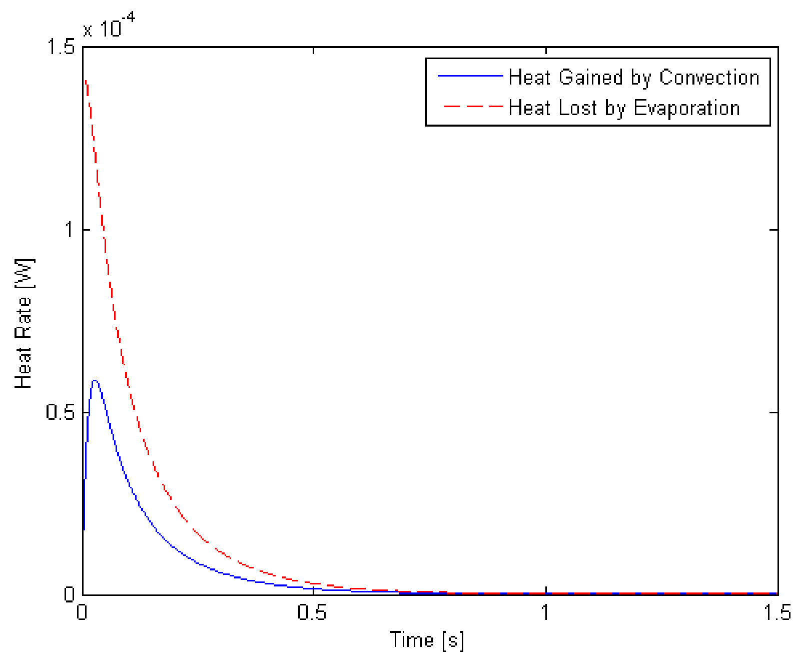

Figure 4. Contributions by convection and latent heat transfer on a droplet starting at 80°F

Initially there is no convection. Evaporation is the only method of heat transfer. The droplet's temperature drops rapidly due to evaporation, allowing convection to take a larger role.

Relative humidity and vapor pressure increase to the identical situation we saw in Case I: convection and latent heat transfer act against each other in a quasi-equilibrium state and slowly bring the droplet temperature down.

Case III: $T_d > T_a$

Figure 5 shows the behavior of the system when the initial water temperature is 140 °F.

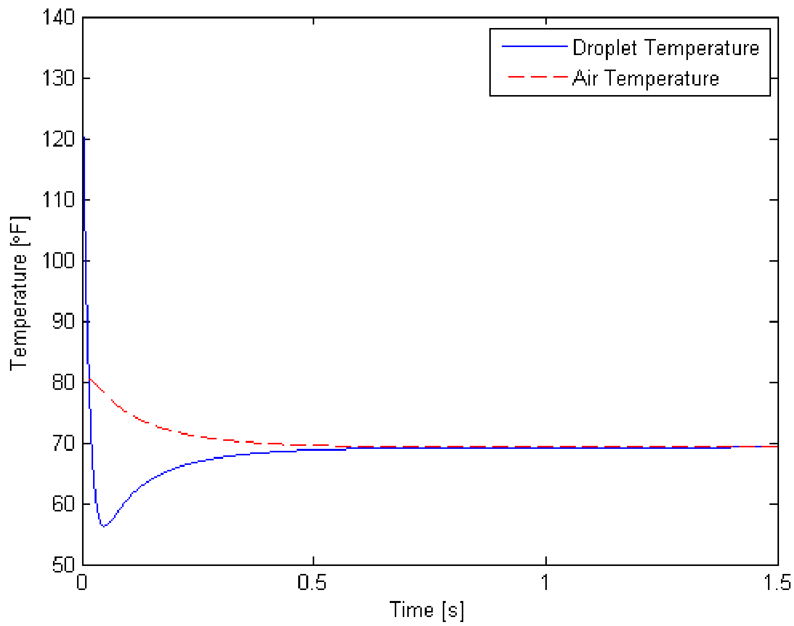

Figure 5. Transient behavior of 80°F air and a 140°F droplet at 50% RH

Figure 6 presents a graph of $\dot{Q}_{conv}$ and $\dot{Q}_{lat}$ to show how convection and latent heat transfer contribute to the temperature changes taking place in the droplet. In conjunction with Figure 5, it can explain how heat and mass transfer between the droplet and air.

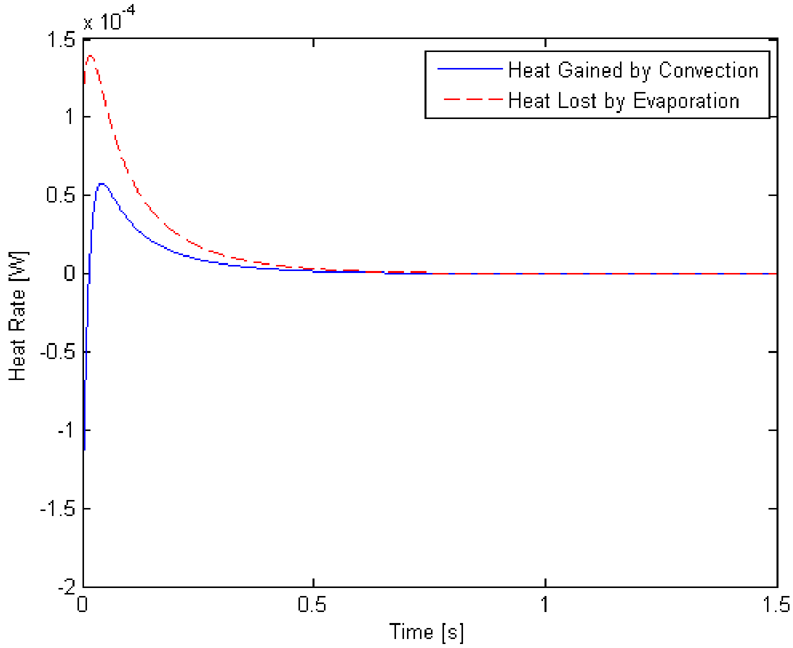

Figure 6. Contributions by convection and latent heat transfer on a droplet starting at 140°F

As the droplet exits the nozzle, convection dominates. The droplet is 60 °F warmer than the air, so the droplet is convecting heat to the air. The air temperature actually goes up 1 °F. This is the first case in which there is a source to heat the air.

The droplet temperature drops rapidly from convection and evaporation. When the droplet temperature drops below the air temperature, convection becomes positive. From then on, we see a quasi-equilibrium state similar to the previous cases.

Case Comparison

The previous three-case explanation failed to include a discussion on relative humidity and droplet diameter. Figures 7 and 8 present these comparisons. In each of these graphs, the view is zoomed in accentuate the contrast between cases. The time scale only shows the first 0.75 seconds, rather than the entire simulation.

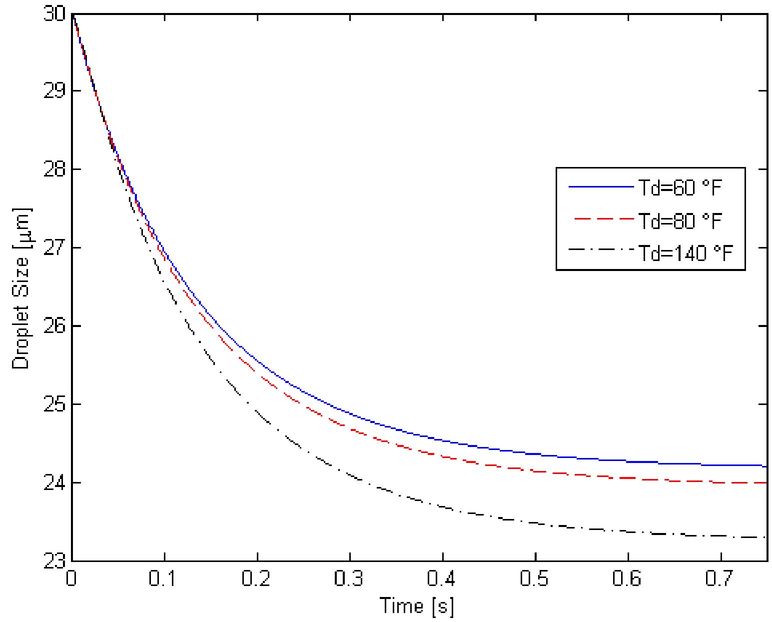

Figure 7. Droplet diameter vs. time for 30 micron droplets starting at 60°F, 80°F, and 140°F

Final droplet diameter decreases as initial temperature increases. Higher temperatures yield higher energy particles that more readily evaporate. However, because the heat of vaporization decreases as temperatures increase, particles that evaporate at higher temperature are worth less in terms of overall heating. Therefore, the overall effect of higher droplet temperature is less evaporative cooling per unit volume of evaporated vapor.

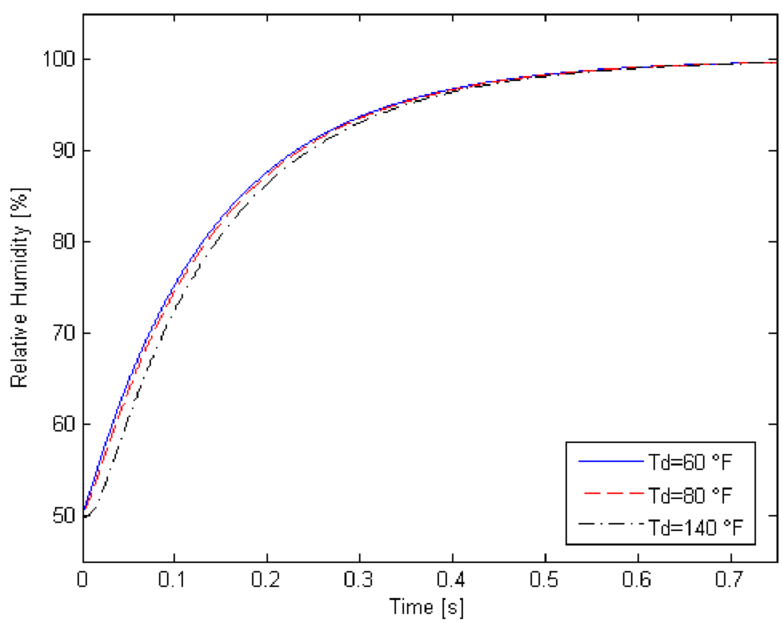

Figure 8. Relative humidity vs. time for 30 micron droplets starting at 60°F, 80°F, and 140°F

In all cases, relative humidity increases to almost 100%. In the 140 °F case, note the initial delay. As the droplet heats the heats the air, relative humidity actually decreases. Once the droplet temperature falls below the air temperature, relative humidity will begin to increase.

Running the same simulation for initial droplet temperatures from 60 to 200 °F, one can understand the effects of inlet water temperature on final air temperature. Also presented are the final droplet diameter and the density increase from ambient to the filter house exit.

Table 2. Simulation Results for droplet temperatures 60 °F to 200 °F and initial air temperature 80 °F

Initial Water Temperature [°F]	Final Air Temperature [°F]	Finial Droplet Diameter [μm]	Air Density Increase
60.0	68.20	24.17	1.84%
80.0	68.45	23.95	1.78%
100.0	68.70	23.72	1.72%
120.0	68.95	23.49	1.66%
140.0	69.21	23.25	1.60%
160.0	69.48	23.01	1.54%
180.0	69.75	22.77	1.47%
200.0	70.03	22.53	1.41%

As initial droplet temperatures increase, final air temperatures increase, regardless of evaporation rates and droplet sizes. However, this temperature change is small. Droplet temperature must increase 77 °F to change the final air temperature by 1 °F. Also, notice that as droplet temperatures increase, there is less increase in air density. This is in exact opposition to the goal of inlet fogging: to increase air density.

Recall, water vapor is less dense than air. So increasing the amount of evaporated vapor has an adverse effect on overall air density. The percentage increase in mass from evaporation as temperature changes is minute and cannot counteract the density decrease. The mixing ratio in this case is approximately 130 pounds of air per pound of water. The diameter difference of 2 microns from 60 to 200 °F has an infinitesimal effect on mass flow.

Model Validity

To test the validity of the model, initial conditions came from actual fogging systems from Plant A and the other combined cycle power plants Mott MacDonald surveyed. Tables 3 - 7 compare the model results to actual plant data and highlight the differences in final air temperature.

Table 3. Turbine Comparison 1

Property	*Actual*	*Simulated*
Initial Air Temperature	79.6 °F	79.6 °F
Wet Bulb Temperature	68.9 °F	68.9 °F
Water Flow Rate	~26.3 gpm	26.3 gpm
Initial Water Temperature	195 °F	195 °F
Load	98%	98%
Final Air Temperature	**71.7 °F**	**70.76 °F**

Table 4. Turbine Comparison 2

Property	*Actual*	*Simulated*
Initial Air Temperature	80.7 °F	80.7 °F
Wet Bulb Temperature	68.6 °F	68.6 °F
Water Flow Rate	27.1 gpm	27.1 gpm
Initial Water Temperature	212 °F	212 °F
Load	98%	98%
Final Air Temperature	**72.4 °F**	**71.61 °F**

Table 5. Turbine Comparison 3

Property	*Actual*	*Simulated*
Initial Air Temperature	81.6 °F	81.6 °F
Wet Bulb Temperature	69.0 °F	69.0 °F
Water Flow Rate	~25.6 gpm	25.6 gpm
Initial Water Temperature	204 °F	204 °F
Load	98%	98%
Final Air Temperature	**73.0 °F**	**72.90°F**

Table 6. Turbine Comparison 4

Property	*Actual*	*Simulated*
Initial Air Temperature	79.3 °F	79.3 °F
Wet Bulb Temperature	67.6 °F	67.6 °F
Water Flow Rate	~24.4 gpm	24.4 gpm
Initial Water Temperature	204 °F	204 °F
Load	99%	99%
Final Air Temperature	**71.7 °F**	**71.10 °F**

Table 7. Turbine Comparison 5

Property	*Actual*	*Simulated*
Initial Air Temperature	63.6 °F	63.6 °F
Wet Bulb Temperature	53.3 °F	53.3 °F
Water Flow Rate	15.5 gpm	15.5 gpm
Initial Water Temperature	84 °F	84 °F
Load	94%	94%
Final Air Temperature	**56.7 °F**	**57.15 °F**

Based on these comparisons, the model has an average error of 0.8% based on the Fahrenheit temperature scale. The greatest error occurred on Turbine Comparison 1 with 1.3% error; the lowest occurred on Turbine Comparison 3 with 0.1% error. A better verification could be made if we were able to verify the initial droplet size, flow rate, and air flow rate through the turbine for each unit.

These tests assume the nozzles are working properly and producing an average droplet size of 10.5 µ, a spec from Plant A's nozzle documentation. They assume a standard air flow rate at full load of 3900 kpph, which was provided by Mott MacDonald's client. They also assume that all the measuring instruments are working properly. If the fogging water flow meter was not working, the Adjusted Maximum GPM from the control screen was used for water flow rate and is indicated with a '~'.

Conclusions

The study results in the following conclusions:

- The primary goal of inlet fogging is to increase the air density so that the mass flow through the turbine and the total power generated increase. Increasing water temperature does not accomplish this.
- Warmer fogger water will cause more evaporation, but will cool the air less effectively than colder fogger water. Colder fogger water will evaporate to a lesser extent, but will result in a greater decrease in air temperature.
- Water vapor is less dense than dry air, so increased evaporation has an adverse effect on the system. Decreasing evaporation and increasing convective cooling are design functions that reflect the primary goal of inlet fogging better than any alternative. This lends itself to larger droplet sizes and colder water. Many operators measure the effectiveness of their foggers by measuring the run-off from the filter house, which is the water that did not evaporate. A common misconception is that too much run-off equates to poor performance. This is not necessarily the case. Decreasing the droplet size and increasing the water temperature will reduce the run-off, but it will also reduce the fogger performance.
- The model presented is valid and makes valid assumptions. Its accuracy was verified within 0.8% error compared to field data.

In light of these conclusions, it is recommended that fogger systems should use water cooler than the air to enhance turbine output. Fogger water that is warmer than the air will lead to poorer fogger performance when compared to equal or lesser water temperatures. The primary goal of inlet fogging is to increase air density. The best foggers should be designed with that primary goal in mind.

References

[1] Aoki, K. (2003). "*Knudsen Layer for Gas Mixtures*," Journal of Statistical Physics, 2003, Volume 112, Numbers 3-4: 629-655.

[2] Buck, A. L. (1981) "*New Equations for Computing Vapor Pressure and Enhancement Factor*," Journal of Applied Meteorology December 1981, pp 1527-1532.

[3] Chaplin, Martin. (2 July, 2012). Water Structure and Science: Water molecule structure. *Water molecule structure*. Retrieved 10 August, 2012, from http://www.lsbu.ac.uk/water/molecule.html

[4] The Engineering Toolbox. (n.d.). Water Vapor - Specific Heat. *Water Vapor - Specific Heat*. Retrieved 10 August, 2012, from http://www.engineeringtoolbox.com/water-vapor-d_979.html.

[5] Grayzeck, Ed. (17 November, 2010). Earth Fact Sheet: Terrestrial Atmosphere. *Earth Fact Sheet*. Retrieved 10 August, 2012, from http://nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html.

[6] Gusarov, A. V.; Smurov, I. (2002). "*Gas-dynamic boundary conditions of evaporation and condensation: Numerical analysis of the Knudsen layer*". Physics of Fluids, p 4242.

[7] "*Handbook of Chemistry and Physics*". 88th ed. 2007. CRC Press: Boca Raton, FL.

[8] Hilsenrath, et al. (1955). "*Tables of Thermal Properties of Gases*," NBS Circular 564.

[9] Incropera; DeWitt, Bergman, Lavine (2007). "*Fundamentals of Heat and Mass Transfer*," 6th edition. John Wiley & Sons. pp. 260-261.

[10] Kerker, M., Chatferjee, A., Cooke, D. D. (1976). "*Brownian Coagulation of Aerosols in the Transition Regime*," Pure and Applied Chemistry, Volume 48, pp. 457-462. Pergamon Press.

[11] Meher-Homji, C.B., Mee, T.R. (2002) "*Inlet Fogging of Gas Turbine Engines – Part A: Fog Droplet Thermodynamics, Heat Transfer and Practical Considerations,*" ASME Turbo Expo 2002, Amsterdam June 2002. ASME Paper No: 2002-GT-30562.

[12] Rogers, R.R. and Yau, M.K. (1989) "*A Short Course in Cloud Physics*," 3rd edition. Pergamon Press.

[13] Smits, A.J., Dussauge, J.P. (2006) "*Turbulent shear layers in supersonic flow*," Birkhäuser, ISBN 0387261400, pg 46

[14] Yan, Fang (2003). "*Numerical Simulations of High Knudsen Number Gas Flows and Microchannel Electrokinetic Liquid Flows*." Doctoral Thesis, Drexel University, Philadelphia, Pennsylvania, USA.

Appendix A: Symbol Definitions

Table of Constants

Symbol	Name	Value/Units
d_w	Molecular diameter of water [3]	$2.75 * 10^{-10}\ m$
g	Gravitational acceleration [7]	$9.80665\ \frac{m}{s^2}$
k	Boltzmann's constant [7]	$1.3806505 * 10^{-23}\ \frac{J}{K}$
k_w	Thermal conductivity of water at 10 °C [7]	$0.5800\ \frac{W}{m \cdot K}$
l	Turbine load	$l \in (0, 1]$
M_{da}	Molar mass of dry air [5]	$0.0289656\ \frac{kg}{mol}$
M_{water}	Molar mass of water [7]	$0.01801528\ \frac{kg}{mol}$
R	Universal gas constant [7]	$8.314472\ \frac{J}{mol \cdot K}$
T_C	Critical temperature of water [7]	$647.14\ K$
Δt	Time step for simulation	$0.0001\ s$
ϵ_a	Air flow through gas turbine at full load	$3.9 * 10^6\ pph$

Table of Variables

Symbol	Name	Units
C_a	Specific heat of air	$\frac{J}{kg \cdot K}$
C_d	Specific heat of water	$\frac{J}{kg \cdot K}$
C_{da}	Specific heat of dry air	$\frac{J}{kg \cdot K}$
C_{wv}	Specific heat of water vapor	$\frac{J}{kg \cdot K}$
c_{da}	Mass percentage of dry air in humid air	$[-]$
c_v	Mass percentage of water vapor in humid air	$[-]$
ΔC_{mass}	Driving force for mass transfer to the droplet	$\frac{kg}{m^3}$
D_d	Diameter of the droplet	m
Gr_m	Mass Grashof number	$[-]$
Gr_t	Thermal Grashof number	$[-]$
h_{cv}	Convective heat transfer coefficient	$\frac{W}{m^2 \cdot K}$
i	Step in simulation	$i \in \mathbb{N}$

K_{mass}	Mass diffusion coefficient	$\frac{m}{s}$
k_a	Thermal conductivity of air	$\frac{W}{m \cdot K}$
Kn	Knudsen number	$[-]$
L_v	Heat of vaporization at T_a	$\frac{J}{kg}$
m_a	Mass of the humid air	kg
m_d	Mass of the droplet	kg
m_{da}	Mass of dry air in humid air	kg
m_v	Mass of water vapor in humid air	kg
Nu	Nusselt number	$[-]$
P_{da}	Partial Pressure of dry air	Pa
P_{knd}	Saturated vapor pressure at T_d	Pa
P_{sat}	Saturated vapor pressure at T_a	Pa
P_{tot}	Total pressure of humid air	Pa
P_{vap}	Partial pressure of water vapor	Pa
P_{wb}	Saturated vapor pressure at T_{wb}	Pa
Pr	Prandtl number	$[-]$
Q_a	Heat quantity of the humid air	J
Q_a	Heat quantity of the droplet	J
Q_{conv}	Heat transferred via convection	J
Q_{lat}	Energy transferred via mass transfer	J
RH	Relative humidity	$[-]$
S_d	Surface area of droplet	m^2
Sc	Schmidt number	$[-]$
SH	Specific humidity	$[-]$
Sh	Sherwood number	$[-]$
T	Temperature	K
T_a	Temperature of the air	K
T_d	Temperature of the droplet	K
T_{wb}	Wet bulb temperature	K
t	Time	s
Δv	Change in specific volume after a phase change	$\frac{m^3}{kg}$
β	Thermal dilation coefficient	K^{-1}
β_m	Mass dilation coefficient	$\frac{m^3}{kg}$
δ_a	Mass diffusivity for air	$\frac{m^2}{s}$
λ	Mean free path of water vapor	m
ϵ_w	Flow of water from fogging skid	gpm
Φ_{evap}	Mass flux to the droplet	$\frac{kg}{m^2 \cdot K}$

σ	Surface tension at droplet's surface	Pa
ρ	Density of humid air	$\frac{kg}{m^3}$
ρ_{da}	Density of dry air in humid air	$\frac{kg}{m^3}$
ρ_{knd}	Density of water vapor at the Knudsen layer	$\frac{kg}{m^3}$
ρ_w	Density of water	$\frac{kg}{m^3}$
ρ_{wv}	Density of water vapor in humid air	$\frac{kg}{m^3}$
μ_a	Dynamic viscosity of air	$\frac{kg}{m \cdot s}$
ϑ	State space	$[-]$

Appendix B: Empirical Relation Statistics

Density of Dry Air

$$\rho_{da} = 2.3723 - 0.00397 * T_a$$

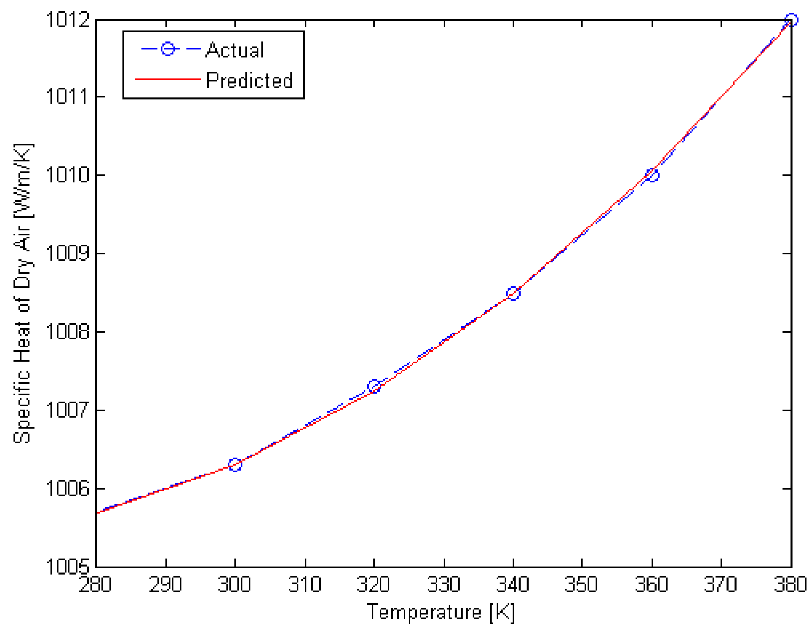

Figure 9. Density of air regression. $R^2 = 0.9965$

Density of Water

$$\rho_w = 786.5 + 1.681 * T_d - 0.003272 * {T_d}^2$$

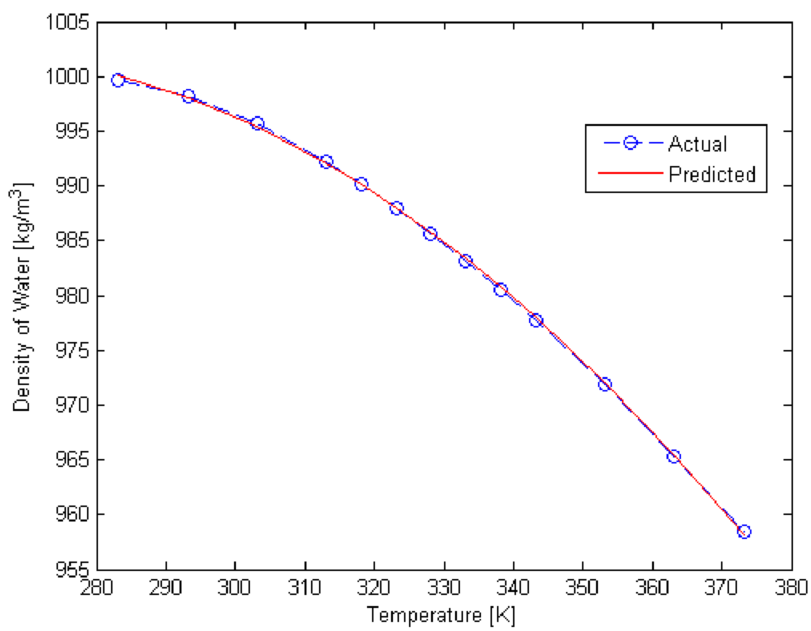

Figure 10. Density of water regression. $R^2 = 0.9997$

Specific Heat of Dry Air

$$C_{da} = 0.000402 * T_a{}^2 - 0.2026 * T_a + 1030.9$$

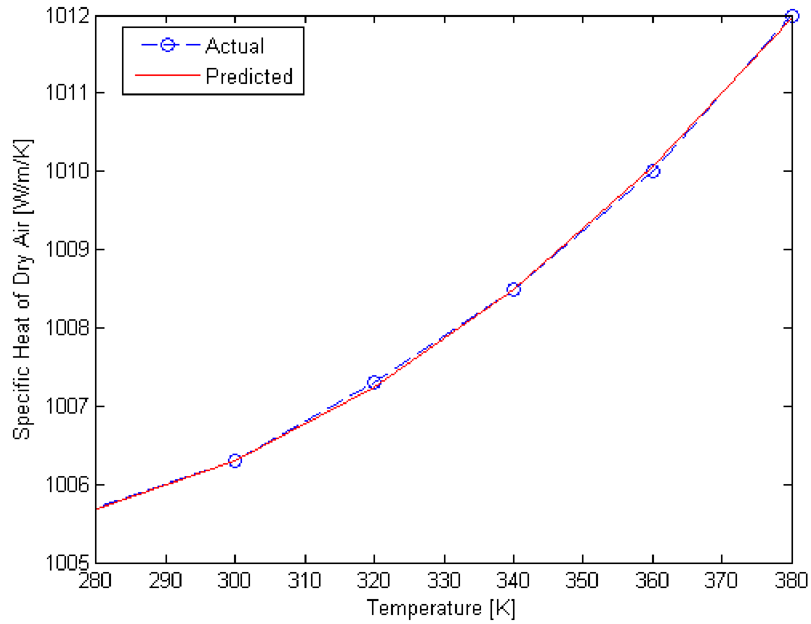

Figure 11. Specific heat of dry air regression. $R^2 = 0.9996$

Specific Heat of Water

$$C_d = -0.000099 * T_d{}^3 + 0.10913 * T_d{}^2 - 39.178 * T_d + 8785.4$$

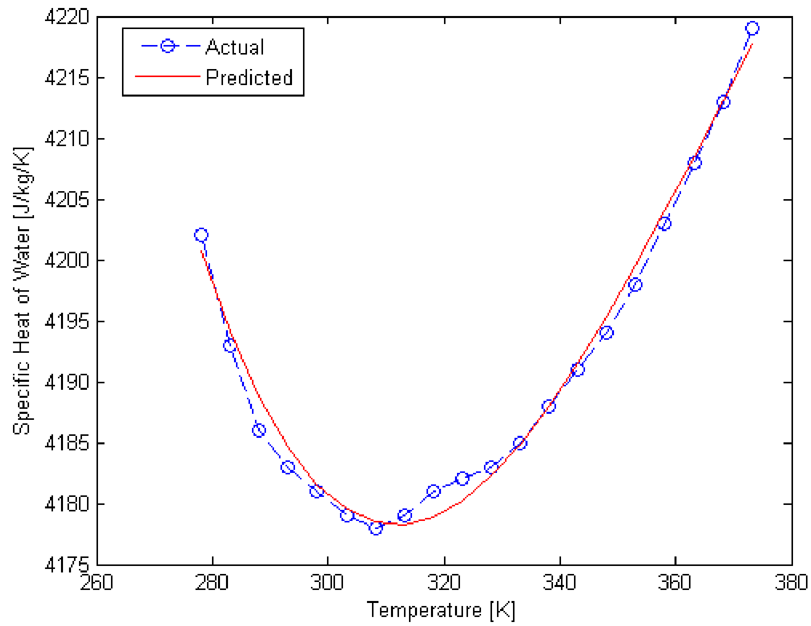

Figure 12. Specific heat of water regression. $R^2 = 0.9893$

Specific Heat of Water Vapor

$$C_{wv} = 1772 + 0.312 * T_a$$

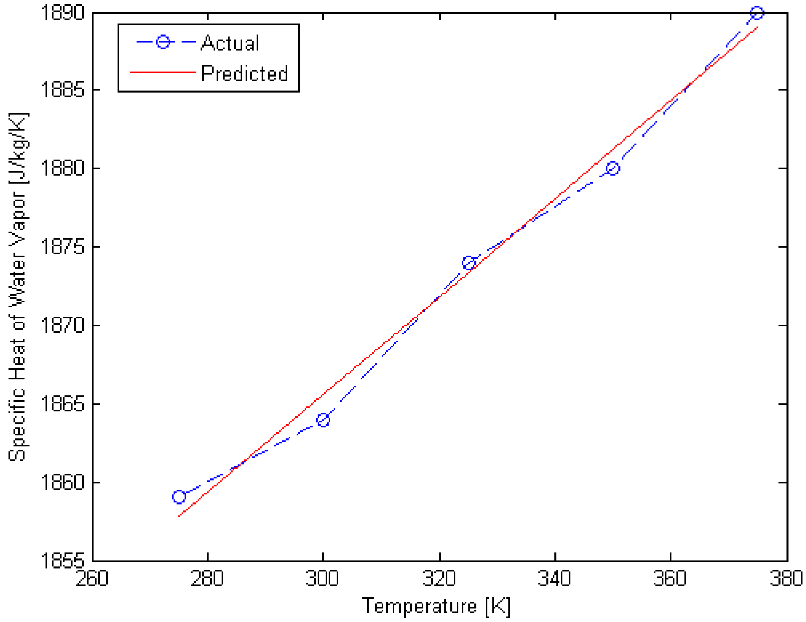

Figure 13. Specific heat of water vapor regression. $R^2 = 0.9889$